Dell PowerFlex 3.6.5 Release Notes

PowerFlex 3.6.5

Rev. 1.1
April 2025
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DELL Technologies

Revision history

Revision history for PowerFlex3.6.5 release notes

Table 1. Revisions

Revision	Date	Description
1.1	April 28, 2025	Technical updates
1.0	February 14, 2025	Release of PowerFlex 3.6.5

Product description

PowerFlex is a software-defined storage platform designed to reduce operational and infrastructure complexity. It empowers organizations to move faster by delivering flexibility, elasticity, and simplicity with predictable performance and resiliency at scale.

The PowerFlex family provides a foundation that combines compute and high-performance storage resources in a managed unified fabric. PowerFlex comes in fully managed flexible deployment options for rack or appliance that enable Server SAN, HCI, and storage only architectures. PowerFlex can now be deployed in the public cloud. A custom node option is also available. PowerFlex is ideal for high-performance applications and databases, building an open private cloud, or consolidating resources in heterogeneous environments.

New and changed features

Learn about new and changed features in this version of PowerFlex.

New features for PowerFlex

Learn about the new features introduced in PowerFlex 3.6.5 software.

- Storage Data Client self-compiles for existing operating systems—In PowerFlex 3.6.3, an SDC self-compile capability is enabled that enables creation of a driver kernel to match the operating system. This process allows customers who want to update the operating system and compile the SDC driver to match the kernel version of the operating system. For more information, see PowerFlex: On-Demand Compilation Of The PowerFlex SDC Driver.

Important

The following aspects of PowerFlex 3.6.5 must be understood before deploying or upgrading your system:

1. CloudIQ integration
   - Discovering a PowerFlex system in CloudIQ requires the system to be registered with Secure remote services. In the PowerFlex Installer, **Maintain** tab, ensure that the PowerFlex system is registered with Secure remote services and that **Send data to CloudIQ** is enabled.
   - After installing a *new* PowerFlex system with PowerFlex version 3.6, it takes 24 hours until data is available in CloudIQ. Systems *upgrading* to PowerFlex 3.6 appears in CloudIQ within an hour of enabling the **Send data to CloudIQ** option.
   - PowerFlex appliance and rack systems need PowerFlex Manager version "H" in order to be discovered in CloudIQ.
2. Replication enhancements
   - Bi-directional replication between two sites only is supported.
   - See the PowerFlex Sizer for sizing guidelines for correct replication system specifications, including WAN bandwidth sizing. When using the PowerFlex Sizer, consider existing workload and existing hardware, and the possible need to add additional nodes to enable replication without impacting the current workload.
   - Replication capability has higher I/O overhead, as explained in the v3.6 *Getting to Know PowerFlex Guide* (Replication section) and the white paper Dell EMC PowerFlex: Introduction to Replication. See these recommendations in order to size your system correctly, and ensure that you have sufficient existing storage capabilities to enable this ability.
3. Network resiliency improvements
   - Proactive network path error detection requires SDC to be at v3.6 or later.

4. Upgrade Paths
   - Upgrade from PowerFlex version 3.0.x and 3.5.x to version 3.6 are supported.
   - Older PowerFlex/ VxFlex operating system versions must be upgraded to version 3.0.x before upgrading to version 3.6.
5. vSphere plug-in End-of-Support-Life
   - The PowerFlex vSphere plug-in continues to be supported with vSphere versions 6.5 and 6.7, however it reaches end-of-support-life when vSphere 6.5 and 6.7 reach end-of-support by VMware. For details on the last release of the vSphere plug-in due to end of general support for ESXi 6.5 and 6.7 by VMware, see: Announcing End of General Support for vSphere 6.5.x and vSphere 6.7.x (89305). Also note the new issue in the *Known Issues* section that is related to vSphere Plug-in.
   - The PowerFlex vSphere plug-in does not support vSphere 7 hyperconverged (HCI) configurations.

Changed features

Learn about the enhanced features for this version of PowerFlex.

Updated operating system support matrix

The following operating systems are added to the support matrix in this release:

- Red Hat Enterprise Linux 8.10
- Red Hat Enterprise Linux 9.3
- Red Hat Enterprise Linux 9.4
- Red Hat Enterprise Linux 9.5
- SLES 15 SP6 for PowerFlex software
- Rocky Linux 8.9
- Rocky Linux 8.10
- Rocky Linux 9.3
- Rocky Linux 9.4
- Rocky Linux 9.5
- AlmaLinux 8.9
- AlmaLinux 8.10
- AlmaLinux 9.3
- AlmaLinux 9.4
- AlmaLinux 9.5
- Ubuntu 24.04
- Debian 12.5
- Windows Server 2025

NOTE: Proxmox VE version 8.2.x based on Debian 12.5 is now supported. You must install SDC using the self-compile feature. For details, see PowerFlex: On-Demand Compilation Of The PowerFlex SDC Driver.

NOTE: Any operating system that is supported with Fine Granularity supports SDPM/NVDIMM.

The operating systems support software only customers for the following components:

- MDM
- SDS
- SDR
- SDC
- LIA

Further to SDC only, the following operating systems support are being added:

- VMware ESXi 8.0 Update 3 (build 24022510)
- VMware ESXi 7.0 Update 3q (build 23794027)

For more operating system-related information and supported components, see the PowerFlex simple support matrix: DELL TECHNOLOGIES SIMPLE SUPPORT MATRICES

Oracle JRE and OpenJDK support

PowerFlex 3.6.5 supports OpenJDK 17. It does not support OpenJDK 11 or 1.8 versions due to end of support for both versions.

Support deprecation

Learn about the features that are no longer supported in PowerFlex 3.6.5.

Background scanner events and alerts

Some events and alerts for the background scanner feature were deprecated, and replaced with different ones. See the *PowerFlex v3.5.1.2 Release Notes*, "Documentation" section for details.

DAS Cache

DAS Cache is no longer supported for PowerFlex running on any operating systems.

System Analysis Tool

The System Analysis Tool is no longer included in PowerFlex.

AMS

Systems that the AMS system manages cannot be upgraded to PowerFlex v3.5 or later. Before performing an upgrade, contact PowerFlex Product Management for possible alternatives.

NOTE: AMS artifacts are removed from the support site. For guidance, contact PowerFlex Product Management.

VxFlex OS UI client

The UI from versions earlier than v3.5 of this product is no longer supported and cannot be upgraded. From v3.5 and later, a new UI must be deployed or upgraded instead.

Deprecated operating systems

For the most current information, see the *Dell Technologies Simple Support Matrix* at DELL TECHNOLOGIES SIMPLE SUPPORT MATRICES.

Fixed issues

The following table lists the issues that are fixed in PowerFlex3.6.5.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and correlate between customer-reported issues and the PowerFlex issue number.

Table 2. Fixed issues

Issue number & SR number	Problem summary
SCI-160656	Expired certificate in containerized version of storage data client 3.6.4
SCI-154509	Ansible is nonresponsive when collecting the information about a disk that is connected to systems running Red Hat Enterprise Linux 9.3 or 9.4. For more information, see DTA

Table 2. Fixed issues (continued)

Issue number & SR number	Problem summary
SR# 196000084	228653: PowerFlex: SCSI command stops responding when collecting information about disk connected to RHEL 9.3 or 9.4 systems.
SCI-152197	Error message displayed when running `showevents` script over Red Hat Enterprise Linux version 8 node using python 3.12
SCI-151681 SR# 193632142	Server 3.6.2.0 Gateway REST API timeout. An Update to libraries in the PowerFlex gateway (CryptoJ) resolves this issue
SCI-156860 SR# 197804031	Gateway times out leading to CSI cluster impact. An Update to libraries in the PowerFlex gateway (CryptoJ) resolves this issue
SCI-157094 SR# 197301015	API Calls from CSI driver fail https handler stopped responding. An Update to libraries in the PowerFlex gateway (CryptoJ) resolves this issue.
SCI-152140 194469902	Storage data client panics with `drl_IsClean`
SCI-68154	Sdc-monitor consumes excessive CPU resources on all Storage Data Clients.
SCI-67273	Windows cannot run `drv_cfg.exe`
SCI-66895 139067588	Storage data client monitor consuming large amounts of CPU resources on all storage data clients as the container uses supervisord to run dmesg and log it. The dmesg switch is not supported by all kernels.

Software packages

Learn about the contents of the PowerFlex 3.6.5 software packages.

Download location

The PowerFlex 3.6.5 complete software package, containing all the packages that are listed below, can be downloaded from the following location:

PowerFlex Software

Table 3. Software packages

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
Red Hat Enterprise Linux_OEL7	Red Hat Enterprise Linux/CentOS/Oracle Linux	X	X	X	X	X	X	X	Core components for Red Hat Enterprise Linux, CentOS, and Oracle Linux 7.x. Oracle Linux 7.8 and higher includes KVM.
Red Hat Enterprise Linux 8_OEL8	Red Hat Enterprise Linux/CentOS/Oracle Linux	X	X	X	X	X	X	X	Core components for Red Hat Enterprise Linux, CentOS, and Oracle Linux 8.x. Oracle Linux includes KVM.
Red Hat Enterprise Linux_OEL9	Red Hat Enterprise Linux/CentOS/Oracle Linux	-	X	X	X	X	X	-	Core components for Red Hat Enterprise

Table 3. Software packages (continued)

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
									Linux, CentOS, and Oracle Linux 9.x.
CENTOS 7.9	-	-	-	-	-	-	-	-	-
CentOS Stream 8	-	X	X	X	X	X	X	X	Core components for CentOS Stream 8
CentOS Stream 9	-	-	X	X	X	X	X	-	Core components for CentOS Stream 9
SLES12.5	SUSE Linux	X	X	X	X	X	X	X	Core components for SLES 12 SP5
SLES15.3	SUSE Linux	X	X	X	X	X	X	X	Core components for SLES 15 SP3
SLES15.4	SUSE Linux	-	X	X	X	X	X	-	Core components for SLES 15 SP4
SLES15.5	SUSE Linux	-	X	X	X	X	X	-	Core components for SLES 15 SP5
SLES 15.6	SUSE Linux	-	X	X	X	X	X	-	Core components for SLES 15 SP6
Ubuntu18.04	Ubuntu	X	X	X	X	X	X	X	Core components for Ubuntu 18.04
Ubuntu20.04	Ubuntu	X	X	X	X	X	X	X	Core components for Ubuntu 20.04
Ubuntu22.04	Ubuntu	-	X	X	X	X	X	-	Core components for Ubuntu 22.04.x LTS
Ubuntu24.04	Ubuntu	-	X	X	X	X	X	-	Core components for Ubuntu 24.04.x LTS
Complete_VMware_SW	vSphere	X	X	X	X	X	-	-	Contains all PowerFlex, VMware-related software artifacts (OVA file, drv_cfg file, SVM updating, SDC package, vSphere plug-in, and VASA provider) **NOTE:** PowerFlex vSphere plug-in is not compatible with ESXi 7.0 and later.
Debian 12.5	-	-	X	X	X	X	X	-	-
Complete_Windows_SW	Windows Server	X	-	-	-	X	X	-	Contains all Microsoft Windows (2012-2019) server-related software artifacts, including the gateway and CSV deployment templates.

Table 3. Software packages (continued)

Package Name	Operating System	GW	MDM	SDS	SDR	SDC	LIA	Xcache	Content
Red Hat Enterprise Linux CoreOS	CoreOS	-	-	-	-	X	-	-	SDC component for Red Hat Enterprise Linux CoreOS
CoreOS		-	-	-	-	X	-	-	SDC component for CoreOS
AIX7.2	AIX	-	-	-	-	X	-	-	SDC component for IBM AIX 7.2
Presentation_Server	-	-	-	-	-	-	-	-	Contains PowerFlex WebUI server (Linux and Ubuntu)
Gateway_for_Linux	-	X	-	-	-	-	-	-	PowerFlex Gateway, which includes the following: Installer, Gateway (for REST API, Alerts, SNMP/ Secure Remote Services) for Linux/ Ubuntu (can be used to deploy PowerFlex on Linux-based operating systems only). CSV deployment templates are also included.
GPG-RPM-KEY	-	-	-	-	-	-	-	-	Contains RPM key (RSA for Linux, DSA for Xen) to authenticate the RPM packages using manual install/upgrade procedure.

NOTE:

- See the latest PowerFlex simple support matrix at DELL TECHNOLOGIES SIMPLE SUPPORT MATRICES
- PowerFlex version 3.5 and later provides a Web UI. User interface packages are no longer available.

Install and upgrade

PowerFlex 3.6.5 can be installed on a clean system or used as an upgrade. Learn important information about installing or upgrading to this version.

Installation

For complete instructions for installing PowerFlex, see the *Deploy PowerFlex Guide*.

Licensing

PowerFlex use requires a license. Call your Dell Technologies account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

Table 4. Upgrade support matrix

ScaleIO/VxFlex OS/ PowerFlex base version	Upgrade path	Supporting documentation
v2.0.1.4, v2.6.x	Two-step upgrade flow: 1. Upgrade to the latest version of v3.0.1.x 2. Upgrade to v3.6.0.x	*Upgrade VxFlex OS to v3.0.x Guide* *Upgrade PowerFlex to v3.6 Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x, v3.6, v3.6.0.x	Single-step upgrade from: • VxFlex OS v3.0/v3.0.0.x/v3.0.1.x • PowerFlex v3.5.x • PowerFlex v3.6.0.x to PowerFlex v3.6.5	*Upgrade PowerFlex to v3.6 Guide*

NOTE: Upgrades from 3.6.4 and later to 4.5.x or 4.6.x versions require a minimum core version of 4.5.3 (to be released soon). Contact your support team if you require guidance.

NOTE: You must upgrade to Java17 to run the current gateway.

CloudLink support

Table 5. CloudLink versions

PowerFlex version	CloudLink version	Description of change
3.6.4 and later	8.1.1	Existing CloudLink install base with the original device path logic: /dev/sd*X*.
3.6.0.6 and later	7.1.9	Existing CloudLink install base with the original device path logic: /dev/sd*X*.
3.6.0.2, 3.6.0.3, 3.6.04, and 3.6.0.5	7.1.3	CloudLink 7.1.3 enables a new device path logic for encrypted devices, using /dev/disk/by-id/wwn-*<unique_id>*.
3.0.1.x all versions 3.5.x.x, all versions 3.6 3.6.0.1	7.1 7.0	The CloudLink versions maintain legacy support for the existing CloudLink install base with the original device path logic: /dev/sd*X*.

PowerFlex 3.6.0.2 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.2 or CloudLink 7.1.3.

Following are the expected upgrade flow for systems with CloudLink:

- Upgrade PowerFlex to 3.6.0.6.
- For CloudLink 7.1 / 7.0, upgrade to CloudLink 7.1.5.
- For CloudLink 7.1.2 / 7.1.3 / 7.1.4, upgrade to CloudLink 7.1.5 with support assistance.

PowerFlex 3.6.0.6 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.5.

CloudLink 7.1.5 changes the disk-mapping logic to use /dev/sd*X* compared to what was introduced in the CloudLink 7.1.2 to 7.1.4 versions. This version removes support for /dev/disk/by– id from CloudLink which reverts to a /dev/sd*X* device mapping.

Customers who migrated to CloudLink 7.1.3 and adopted `/dev/disk/by-id` logic must open a support call as documented in *Dell PowerFlex v3.6.x CloudLink for SDS Devices*. This helps in reverting to /dev/sd*X* device mapping and helps adopt the CloudLink 7.1.5.

Customers using the earlier CloudLink releases (CloudLink 7.1.1 or earlier) can adopt the CloudLink 7.1.5 by using the existing device mapping.

Due to key updates, it is recommended that you adopt the latest CloudLink 7.1.5 for usage with PowerFlex for Data at Rest Encryption.

Supported operating systems

The user documentation contains a list of supported operating systems. For the most current information, see the *Dell Technologies Simple Support Matrices* at DELL TECHNOLOGIES SIMPLE SUPPORT MATRICES, **Storage** option.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with previously released systems that are supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward compatible with previously released SDC versions that are supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- You cannot enable the CHAP component authentication feature unless every MDM, SDC, and SDS in the system is at least version 3.5. If the feature is enabled, you cannot add any pre-v3.5 SDCs.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex operating system v2.x SDC.
- Snapshot Policy management is not available with SDC releases earlier than VxFlex operating system v3.0.

Known issues and limitations

The following table lists known issues and limitations that exist in PowerFlex 3.6.5.

NOTE: If an issue was reported by customers, the Service Request numbers of the customers appear in the "Issue number & SR number" column, and serve to correlate between customer-reported issues and the PowerFlex issue number.

Table 6. Known issues and limitations—PowerFlex Gateway

Issue number & SR number	Problem summary	Workaround
SCI-66203 SR# v989309	The AutoComplete attribute is not disabled for passwords in form-based authentication. If the browser is used in a shared computing environment where more than one person may use the browser, "autocomplete" values may be retrieved or submitted by an unauthorized user.	None
SCI-58953	When running the PowerFlex Gateway on an Internet Explorer 11 browser, LIA Authentication gets stuck in "Loading..." status when retrieving system topology.	Use Chrome or Firefox browsers to perform this action.
SCI-58471	PowerFlex Gateway Tomcat log rotations do not work after an upgrade. At build time, the "server.xml" file is defined as a "noreplace-config file" in the RPM. During an upgrade, the RPM mechanism will automatically detect if the file was previously edited or not. If the file was edited, it will not be overwritten during upgrade, and the new file will be created with the ".rpmnew" suffix.	Edit the file "server.xml" manually, and change the log rotation rules. For more information, see the PowerFlex Troubleshooting Guide, "Some PowerFlex Gateway log files fail to rotate" topic.
SCI-57810	The PowerFlex Gateway Maintenance operation screen might show an invalid LIA authentication for a containerized SDC when connecting to the system. This should appear as N/A, and can be ignored.	None

Table 6. Known issues and limitations—PowerFlex Gateway (continued)

Issue number & SR number	Problem summary	Workaround
SCI-57304	PowerFlex Installer (also known as the PowerFlex Gateway) is exposed to issue HSTS Missing From HTTPS Server, CVE: CVE-2017-7789 Detail	None
SCI-54887	When performing the log collection using the PowerFlex Gateway with the "exception only" check box marked, the log collection fails.	Perform log collection with the default settings using the PowerFlex Gateway.
SCI-54799	If Auto Collect logs is configured in the system during an NDU session, when the NDU ends, the PowerFlex Installer displays a button asking to start the first phase of the NDU, even though the previous one was completed successfully (instead of displaying the "Mark operation as done" button).	Disable auto log collection prior to starting the NDU, and run the NDU. After NDU completion, enable auto log collection again.

Table 7. Known issues and limitations—MDM

Issue number & SR number	Problem summary	Workaround
SCI-60885	When using SRA, the PowerFlex Gateway password and the MDM password must be the same. Otherwise. one of the following errors is displayed on the SRM/SRA UI: SRA command 'discoverArrays' failed. General exception No further information available SRA command 'discoverArrays' failed. general error check logs for detail	Change the PowerFlex Gateway and MDM user passwords simultaneously. Ensure that they have admin permissions and the same password.
SCI-69646	By default, Python3 is installed on Ubuntu and is located in `/usr/bin/python3`. During the execution of the command `/opt/emc/scaleio/mdm/diag/dump_and_balance_check.py --username XXXXX --password YYYYY --approve_certificate --scripts_path=/opt/emc/scaleio/mdm/diag/`, the following error is encountered: `bash: /opt/emc/scaleio/mdm/diag/dump_and_balance_check.py: /usr/bin/python: bad interpreter: No such file or directory.`	Add "3" at the start of the Python script.

Table 8. Known issues and limitations—Presentation server

Issue number & SR number	Problem summary	Workaround
SCI-60813	When connecting a presentation server from PowerFlex version 3.6 to a PowerFlex 3.5 system, some of the UI functionality does not work as expected.	Use the matching version of the presentation server. In an upgrade flow, see the upgrade guide for guidelines about when and how to upgrade the presentation server.

Table 9. Known issues and limitations—SDC

Issue number & SR number	Problem summary	Workaround
SCI-57752	The SDC rpm package for SLES15.X installs without any issues. However, after installation, when checking if the SDC driver's scini service is loaded, the following confusing state is returned: service scini status service: no such service scini This is a false positive status alert.	None
SCI-69559	SDC installed with flash vSphere plug-in fails without any recovery due to end of support by VMware for ESXi 6.7.x and 6.5.x. For details, see Announcing end of general support for vSphere 6.5.x and vSphere 6.7.x (89305).	None

Table 10. Known issues and limitations—vVols

Issue number & SR number	Problem summary	Workaround
SCI-54057	Unmapping a VMware virtual volume, and then immediately afterwards trying to overwrite its content, might generate the error: "volume n is in use".	None. Try again shortly after, because some volume operations are performed in an asynchronous fashion.

Operating system known issues and limitations

Learn about known issues and limitations for specific operating systems.

Table 11. Known issues and limitations—operating systems

Issue number & SR number	Problem summary	Workaround
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-360	The device name appears incorrectly as Non PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 and 6.00.30.202, respectively).	Dell Technologies recommends upgrading the OpenManager Enterprise catalog to the latest version.
—	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest supported version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see 15G PowerFlex Family running CPLD 1.0.7 can hang during POST when updating to BIOS 1.8.2 and iDRAC 6.00.30.202.

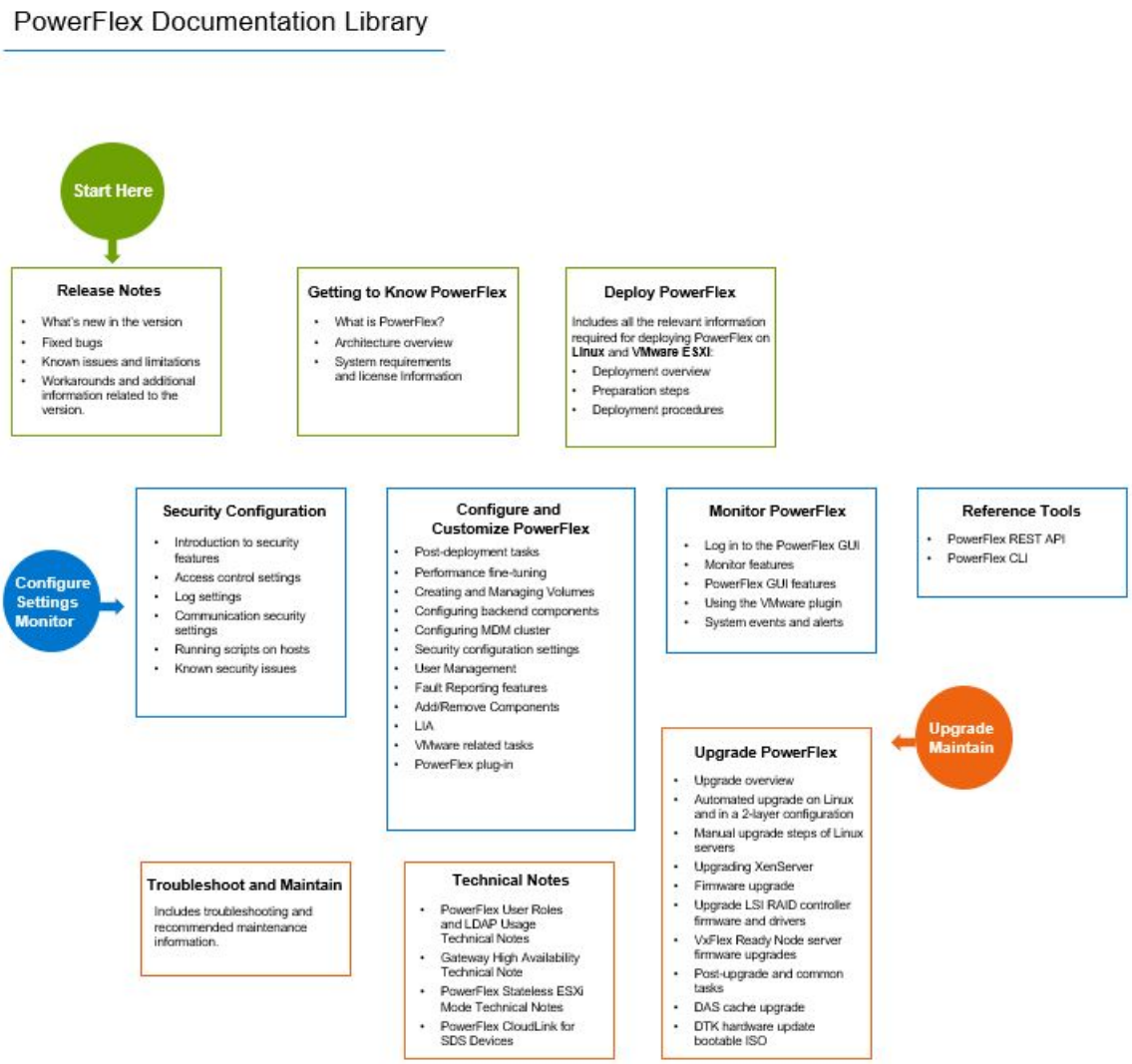
Additional resources

Use these resources to find more information about this product, get support, and provide feedback.

Product information

For documentation, release notes, software updates, or information about Dell products, go to Dell Online Support at:

PowerFlex Software



Where to get support

Go to Dell Online Support and click **Support**. There are several options for contacting Dell Technical Support. Note that to open a service request, you must have a valid support agreement. Contact your Dell sales representative for details about obtaining a valid support agreement or with questions about your account.

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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